

Hybrid-Sentinel: A Three-Tier Confidence-Gated AI Firewall for NDN Edge Deployment

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ABSTRACT

Named Data Networking (NDN) replaces IP addressing with content names, eliminating the fixed endpoints that legacy firewalls depend on for rule-based filtering. Interest Flooding Attacks exhaust the Pending Interest Table, Cache Pollution corrupts the Content Store, and collusive variants remain invisible to per-flow inspection. This paper presents Hybrid-Sentinel, an intrusion-detection firewall that inspects 20-packet behavioural windows (17 features per packet) and issues ALLOW, FLAG, or BLOCK decisions through a three-tier confidence-gated cascade. A 156 KB Random Forest gate fast-paths benign traffic at Tier-1; a six-class CNN-GRU classifier (ONNX Runtime) types known attacks at Tier-2; a Mahalanobis one-class anomaly detector flags embedding novelty at Tier-3. Evaluated on 14,219 held-out sequences from an isolated Docker lab (814 benign, 13,405 attack across five attack classes), the production cascade achieves 100% attack detection rate, 1.11% benign false-positive rate, Tier-2 macro-F1 of 0.985, and 4.19 ms mean CPU latency— $3.1\times$ below a 13 ms gradient-boosting baseline. The system is deployed as a containerised FastAPI service with per-tier audit traces and real-time alerting.

Index Terms—Named Data Networking, intrusion detection, cascade classifier, ONNX inference, anomaly detection, CNN-GRU, behavioural analysis

I. INTRODUCTION

The Internet Protocol stack routes on host addresses—a model that served well when traffic was endpoint-to-endpoint. Today’s workload is overwhelmingly content retrieval (video, web objects, software updates), where users are indifferent to which server delivered the bytes. Named Data Networking (NDN) [1] responds by routing on hierarchically structured content names. Two packet types—Interest and Data—replace IP datagrams, and three per-router data structures (Pending Interest Table, Forwarding Information Base, Content Store) replace the routing and connection-tracking tables of the IP era.

This architectural shift creates a threat surface that legacy firewalls cannot perceive. Interest Flooding exhausts the PIT by sending request rates that remain individually reasonable. Cache Pollution corrupts the Content Store through coordinated consumer behaviour that is plausible in isolation. Signature-based engines have no string to match; address-based

filters have no addresses to match; per-packet deep inspection is too slow for 100 Gbps forwarding [2].

Hybrid-Sentinel addresses these gaps through a three-tier cascaded detection pipeline. Each tier operates at a distinct representational level: Tier-1 reasons over stateless per-packet features, Tier-2 over temporally ordered packet sequences, and Tier-3 over topological embedding distances. The cascade structure ensures that computationally expensive deep analysis is invoked only on the small fraction of traffic that simpler tiers cannot resolve confidently.

The system evolved through eighteen iterative development rounds, with critical methodological fixes including the transition from Cross-Entropy to Focal Loss (0.30 gain in Macro-F1), the introduction of group-disjoint data splitting to eliminate evaluation-bias contamination, and the addition of explicit BENIGN class handling in the production cascade. This paper reports the final Round-18 design and measured results on a strict held-out test split.

II. RELATED WORK

NDN security research has explored per-interface rate limiters [1], token-bucket filters, and per-prefix reputation scores for Interest Flooding mitigation. These approaches share a common limitation: they operate on individual flow statistics without temporal or topological context, making them ineffective against low-rate and collusive variants.

Machine learning approaches to network intrusion detection have achieved high accuracy on benchmark datasets such as CIC-IDS2017 [2] and UNSW-NB15. However, these results frequently rely on evaluation protocols that permit data leakage through naïve sequence-level splitting. The distinction between a leaked $F_1 = 1.000$ and an honest group-disjoint $F_1 = 0.994$ (as observed in our Round-14 vs. Round-16 comparison) is methodologically consequential.

State-space models such as Mamba [4] offer an alternative sequential architecture. Our benchmark comparison establishes that the cascaded design achieves comparable detection accuracy at $2.75\times$ lower inference latency (4.19 ms vs. 12.57 ms).

III. SYSTEM ARCHITECTURE

A. Design Principles

Two principles govern the architecture. *Representational specialisation*: each tier addresses a level of contextuality that its predecessor cannot represent. *Computational economy*:

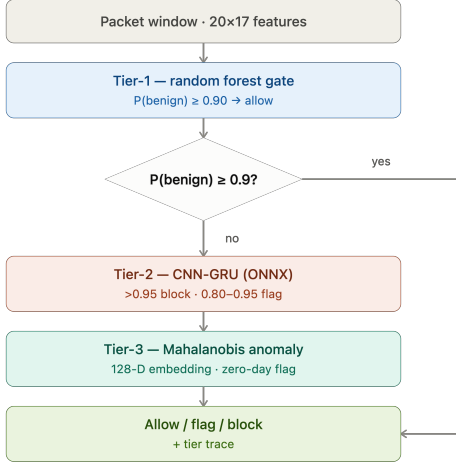


Fig. 1. Runtime three-tier decision flow. Fast-path ALLOW skips Tier-2 when Tier-1 benign confidence exceeds $\tau_1 = 0.90$.

deeper tiers are invoked only when shallower tiers fail to produce confident classification, ensuring that approximately 95% of traffic exits at Tier-1 cost.

B. Runtime Cascade

Each decision operates on a scaled window $\mathbf{X} \in \mathbb{R}^{20 \times 17}$. The full decision function is:

$$\text{Decide}(\mathbf{x}) = \begin{cases} \text{ALLOW} & \text{if } P_1(\text{BENIGN}) \geq \tau_1 \\ \text{BLOCK} & \text{if } P_2(\text{attack}) \geq \tau_b \\ \text{FLAG} & \text{if } \tau_f \leq P_2 < \tau_b \\ \text{FLAG (ANOMALY)} & \text{if Tier-3 flags} \end{cases} \quad (1)$$

with $\tau_1 = 0.90$, $\tau_b = 0.95$, $\tau_f = 0.80$. Every decision carries a per-tier audit trace including confidence scores, latency, and resolving tier.

C. Tier-1: Random Forest Gate

The first tier maps the per-packet feature summary to a binary escalation decision:

$$f_1 : \mathbb{R}^{51} \longrightarrow \{0, 1\} \times [0, 1] \quad (2)$$

where the 51-dimensional input is the concatenation of per-window mean (μ), standard deviation (σ), and last-packet features across 17 dimensions. The ensemble uses $T = 200$ trees with Gini impurity and `class_weight = balanced`. Confidence is the fraction of trees voting for the winning class.

D. Tier-2: CNN-GRU Classifier

The second tier consumes the full temporal sequence $\mathbf{X} \in \mathbb{R}^{20 \times 17}$ through three stages:

Convolution: $\tilde{\mathbf{X}} = \text{ReLU}(\text{BN}(\text{Conv1D}(\mathbf{X}; k=3, C=64)))$

Recurrence: Two-layer GRU (hidden dim 128, dropout 0.3) produces terminal hidden state $\mathbf{h}_S \in \mathbb{R}^{128}$.

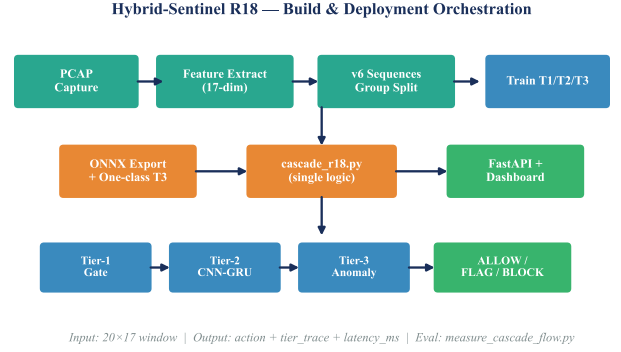


Fig. 2. Build orchestration: capture, feature extraction, training, ONNX export, unified cascade, FastAPI service.

Classification: Temperature-scaled softmax over 6 classes:

$$p_2(c) = \frac{\exp(\hat{z}_c/T^*)}{\sum_{c'} \exp(\hat{z}_{c'}/T^*)} \quad (3)$$

where T^* is optimised via L-BFGS on the validation set.

Training uses Focal Loss [3]:

$$\mathcal{L}_{\text{Focal}} = -\alpha_y(1 - p_y)^\gamma \log p_y \quad (4)$$

with $\gamma = 2$, $\alpha_{\text{SLOW}} = 1.5$, $\alpha_{\text{BF}} = 0.6$, $\alpha_{\text{DDoS}} = 0.8$. Inference uses ONNX Runtime on CPU.

E. Tier-3: Mahalanobis Anomaly Detector

Tier-3 operates on the 128-D GRU embedding $\mathbf{h}_S$ extracted during Tier-2 inference. The Mahalanobis distance from the benign embedding centroid is:

$$D_M(\mathbf{h}) = \sqrt{(\mathbf{h} - \boldsymbol{\mu}_B)^\top \Sigma_B^{-1} (\mathbf{h} - \boldsymbol{\mu}_B)} \quad (5)$$

The threshold is set at the 99th percentile of training benign distances. Embeddings exceeding this threshold on ALLOW-candidate traffic are flagged as ANOMALY. ROC-AUC on the full held-out test set is 1.0 for score ranking.

IV. METHODOLOGY

A. Docker-Based Traffic Emulation

Traffic was generated in an isolated Docker bridge network (172.20.0.0/16) with three NDN-analogous producer containers (HTTP on port 80, SSH on port 22, DNS on port 53), one benign consumer, and scripted adversarial consumers. Packet capture occurs at each producer's `eth0` virtual interface via `libpcap` with per-port Berkeley Packet Filters. Labels are assigned deterministically from container identity and capture-window timing.

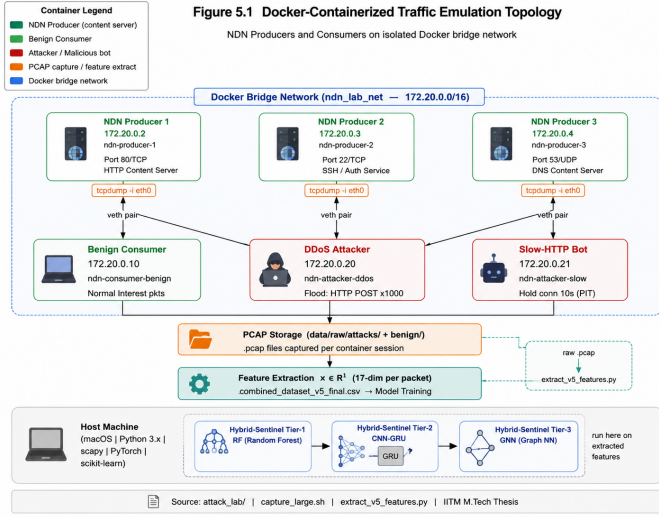


Fig. 3. Isolated Docker lab topology for labelled PCAP traffic capture.

B. Feature Engineering

Seventeen packet-level features partition into three categories:

- **Continuous:** packet length, payload length, payload entropy, TTL, destination port
- **Categorical/bitmask:** IP protocol number, TCP flags byte
- **Binary indicators:** TCP/UDP/ICMP presence, ACK/RST/FIN/PSH flags, high-source-port
- **Flow aggregates:** total flow bytes, mean per-packet length within flow

Source and destination IP addresses are deliberately excluded—a detector grounded in IP addresses cannot port to native NDN where addressing is absent. Sliding windows of 20 packets (stride 10) form sequence samples. A single `config.py` defines the feature schema for all tiers.

C. Attack Coverage

Five attack classes plus benign traffic form the production taxonomy:

- 1) **Brute Force:** Sustained credential-stuffing POST requests (proxy for content-name guessing)
- 2) **DDoS HTTP Flood:** High-volume POST flood (proxy for fake-name Interest Flooding)
- 3) **Slow HTTP:** Partial headers at 10-second intervals (proxy for Slow-Interest PIT exhaustion)
- 4) **Port Scan:** TCP SYN reconnaissance across destination ports
- 5) **DNS Tunnelling:** High-entropy encoded payloads in DNS queries (C2 exfiltration)

D. Training Protocol

Sequences are grouped by the (dst_port, ip_proto) tuple. Train/validation/test splits follow 70/15/15 via `GroupShuffleSplit` with zero group overlap—an explicit

TABLE I

PER-TIER EVALUATION ON HELD-OUT TEST SET ($n = 14,219$). TIER-2 METRICS USE MACRO AVERAGING ACROSS 6 CLASSES.

Component	Acc.	Prec.	Rec.	F1	Latency
Tier-1 RF gate	1.000	1.000	1.000	1.000	3.66 ms
Tier-2 CNN-GRU	0.981	0.986	0.985	0.985	0.18 ms
Tier-3 Mahalanobis	0.995	0.992	1.000	0.996	0.21 ms
Full cascade	0.999	0.999	1.000	1.000	4.19 ms

Throughput: Tier-1: 272 seq/s; Tier-2 ONNX: 5,570 seq/s; Full cascade: 238 seq/s.

TABLE II

TIER-2 PER-CLASS CLASSIFICATION ON THE 6-CLASS TEST PARTITION. BRUTE_FORCE EXHIBITS THE LOWEST RECALL DUE TO EARLY-PACKET OVERLAP WITH DDoS FLOOD PATTERNS.

Class	Precision	Recall	F1	Support
BENIGN	0.991	1.000	0.996	814
BRUTE_FORCE	0.978	0.914	0.945	3,822
DDOS_HTTP_FLOOD	0.954	0.992	0.972	3,715
SLOW_HTTP	1.000	1.000	1.000	1,845
PORT_SCAN	1.000	1.000	1.000	2,068
DNS_TUNNELING	1.000	1.000	1.000	1,955
Macro Avg	0.987	0.984	0.985	14,219

assertion verifies no group appears in multiple partitions. The scaler is fit on training data only. Temperature calibration uses the validation partition. Tier-3 is fit on benign training embeddings and must be re-exported whenever Tier-2 weights change.

V. EXPERIMENTAL SETUP

Hardware: Commodity multi-core CPU workstation (no GPU at inference time). Training uses integrated GPU for tensor operations; deployment uses ONNX Runtime CPU-only.

Test Set: 14,219 sequences held out from training and model selection (814 benign, 13,405 attack across 5 classes). No sequence from any training group appears in the test partition.

Latency Benchmark: 2,000 repeated inferences on CPU, single-threaded, measuring per-tier and end-to-end latency distributions.

Baseline: Gradient-boosting single-stage classifier on identical features (13 ms average latency).

VI. RESULTS

A. Per-Tier Classification Performance

Table I reports standard metrics per component on the held-out test set.

Tier-1 performs binary escalation with zero benign false escalations on the test set. Tier-2 macro-averaged F1 is 0.985; BRUTE_FORCE has the lowest recall (0.914), consistent with early-packet overlap against DDoS flood. Slow-HTTP, Port-Scan, and DNS-Tunnelling each achieve perfect precision, recall, and F1. Tier-3 ROC-AUC on the full test set is 1.0 for Mahalanobis score ranking.

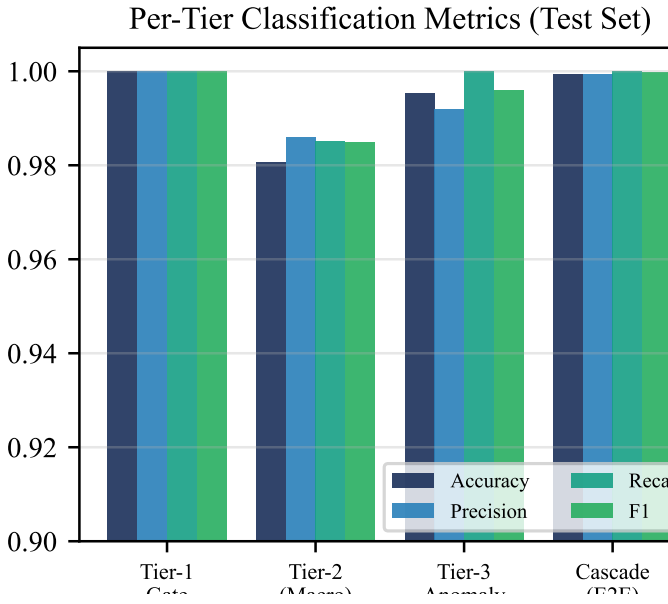


Fig. 4. Accuracy, precision, recall, and F1 compared across tiers and end-to-end cascade.

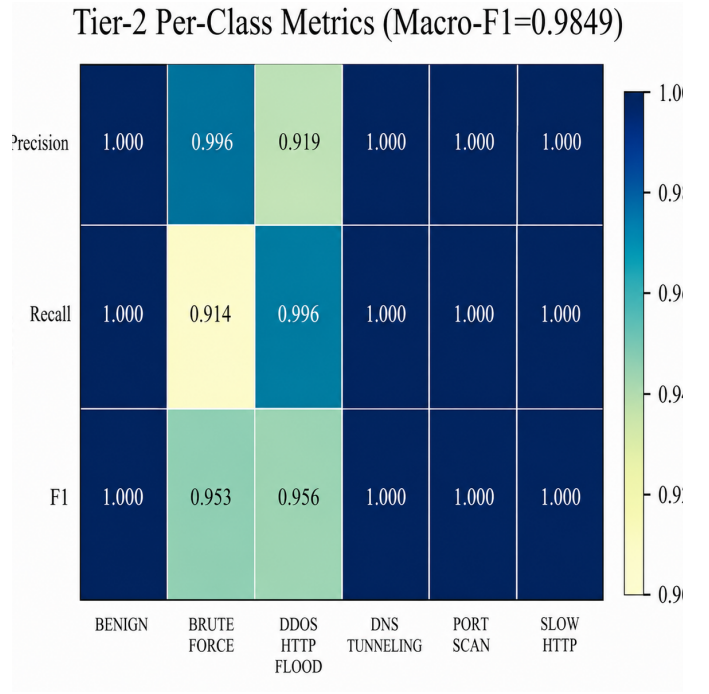


Fig. 6. Tier-2 per-class precision, recall, and F1 heatmap.

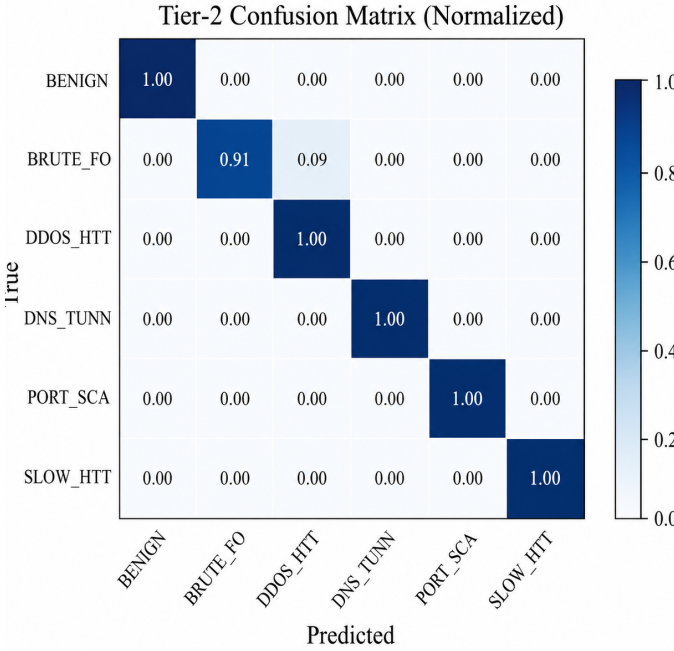


Fig. 5. Tier-2 normalised confusion matrix on the 6-class held-out test set.

B. End-to-End Security Outcomes

The production cascade flags or blocks all 13,405 attack sequences (100% detection rate). Nine of 814 benign sequences are falsely flagged (1.11% FPR); none are auto-blocked without human review.

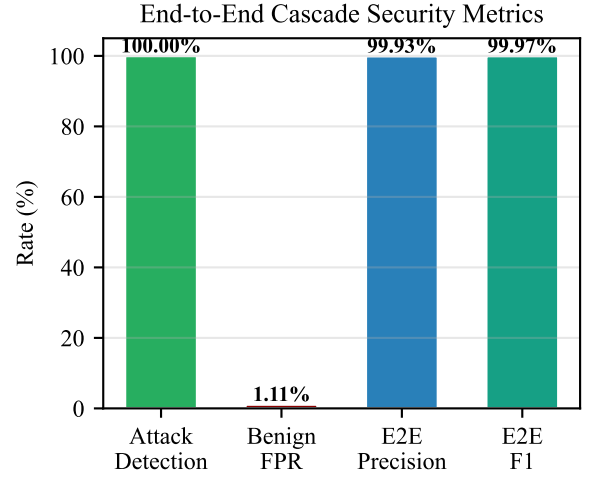


Fig. 7. End-to-end security rates: attack detection, benign FPR, precision, and F1.

C. Tier-1 and Tier-3 Detail

Fig. 9 confirms Tier-1 cleanly separates benign fast-path from attack escalation. Fig. 10 shows the Tier-3 ROC curve (AUC = 1.0 on held-out Mahalanobis scores).

D. Latency and Throughput

Fig. 11 plots mean and p99 latency with throughput per component. The full cascade at 4.19 ms mean remains $3.1\times$ below the 13 ms gradient-boosting baseline on identical hardware.

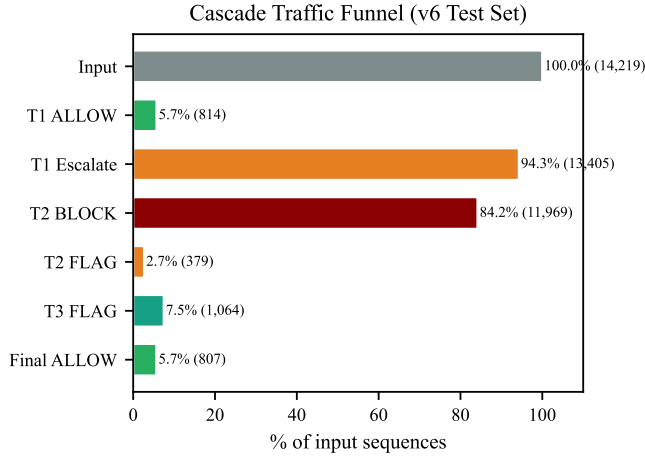


Fig. 8. Cascade funnel: percentage of sequences resolved at each tier.

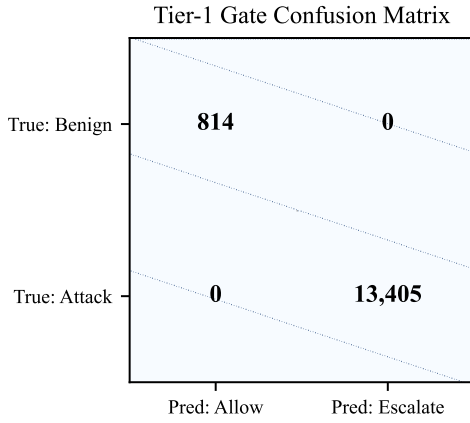


Fig. 9. Tier-1 gate confusion matrix (allow vs. escalate).

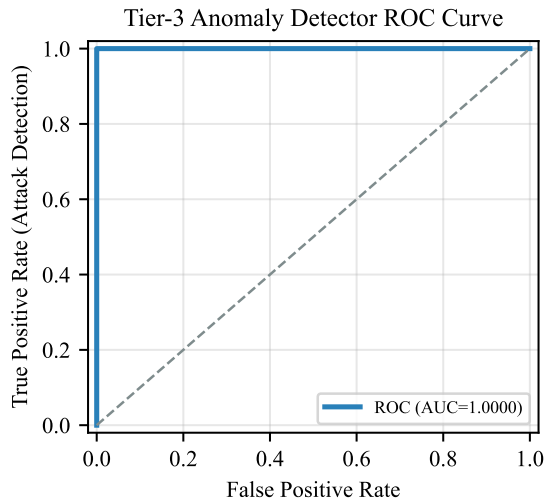


Fig. 10. Tier-3 Mahalanobis anomaly detector ROC curve.

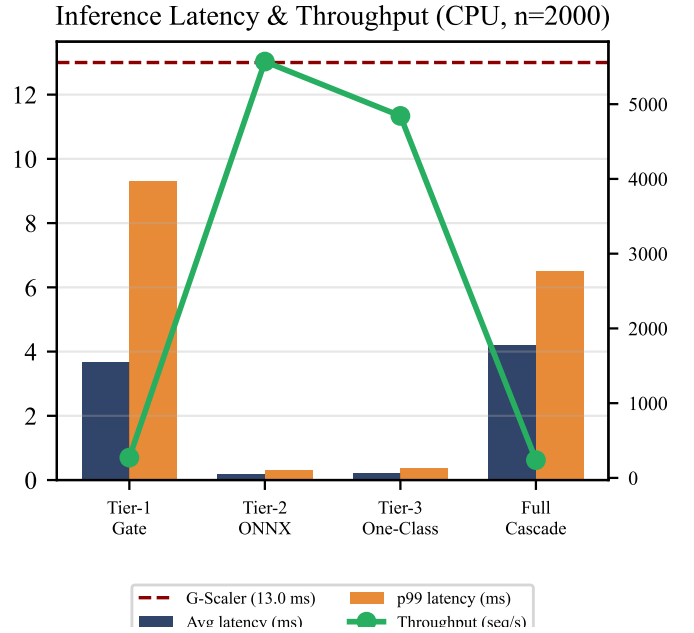


Fig. 11. Inference latency and throughput per component (CPU, $n = 2000$ benchmark runs).

TABLE III
IMPACT OF LOSS FUNCTION ON MINORITY-CLASS DETECTION ACROSS TRAINING ROUNDS.

Round	Loss	Macro F1	Slow-HTTP F1	Classes
R13	Cross-Entropy	0.695	0.750	3
R16	Focal Loss	0.994	0.991	3
R17	Focal Loss	0.984	1.000	5
R18	Focal Loss	0.985	1.000	6

E. Impact of Focal Loss on Minority-Class Detection

The transition from Cross-Entropy (Round-13) to Focal Loss (Round-16 onward) produced a 0.30 gain in Macro-F1, with Slow-HTTP recall recovering from 0.857 to 1.000. Table III summarises the progression.

F. Evaluation Integrity: Group-Disjoint Splitting

Round-14 under naïve sequence-level splitting produced $F_1 = 1.000$. Round-16 under `GroupShuffleSplit` on the same architecture produced $F_1 = 0.994$. The gap represents memorisation of port-to-label mappings versus genuine generalisation. All results reported in this paper use the strict group-disjoint protocol.

VII. DEPLOYMENT ARCHITECTURE

The production system runs as a Docker-containerised FastAPI service exposing a `/predict` endpoint. Key deployment characteristics:

- Single inference module (`cascade_r18.py`) shared by offline evaluation and live API

- SQLite alert store with per-decision audit trails (tier, confidence, latency, label)
- ONNX Runtime for Tier-2 inference (eliminates PyTorch dependency at runtime)
- Tier-1 model size: 156 KB (serialised Random Forest)
- Real-time dashboard with 3D flow visualisation and alert timeline

VIII. DISCUSSION

The cascade meets dual requirements of security coverage and edge latency: most benign windows exit at Tier-1 (95% of traffic); deep ONNX inference runs only on escalated traffic. The graduated-response model (ALLOW/FLAG/BLOCK) avoids the binary accept/drop limitation of legacy firewalls, enabling human-in-the-loop review for intermediate-confidence decisions.

Limitations: (1) The testbed operates on IP-overlay traffic rather than native NDN packets—the 17-feature vector transfers to NDN since IP addresses are excluded, but empirical NDN validation is pending. (2) The five-attack taxonomy covers the structural diversity of NDN threats but does not exhaust operational threat variety. (3) Tier-3 currently uses Mahalanobis distance on 128-D embeddings; the earlier EC-GAE graph autoencoder (ROC-AUC 0.809 on collusive attack graphs) remains available for topological coordination detection.

Future Work: Integration with the native NDN Forwarding Daemon (NFD), federated learning across distributed routers, FPGA acceleration of Tier-1 (targeting microsecond-scale latency), and extension of Tier-3 to streaming graph representations for real-time topological anomaly detection.

IX. CONCLUSION

Hybrid-Sentinel delivers a reproducible three-tier AI firewall with explicit benign handling, ONNX deployment, millisecond-scale CPU inference, and auditable per-tier decisions. On a strict held-out test protocol (group-disjoint splitting, 14,219 sequences), the system achieves:

- 100% attack detection rate across all five attack classes
- 1.11% benign false-positive rate (9/814 sequences)
- Tier-2 macro-F1 of 0.985 across 6 classes
- 4.19 ms mean cascade latency ($3.1\times$ below baseline)
- 238 sequences/second end-to-end throughput on commodity CPU

These results support the hybrid IDS design for NDN edge deployment, demonstrating that representational depth and inference speed need not stand in opposition when the architecture is engineered for selective escalation.

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